

BSCS FINAL PROJECT

<Software Design Specification>

<Vistelligence>



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Software Design Specification

SDP Phase II

<Vistelligence>

Advisor: <Ms Hira Asim>

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Table of Contents

Abstract	1
1. Introduction.....	1
2. Product	2
2.1 Background	2
2.2 Objective(s)/Aim(s)/Target(s)	2
2.3 Scope.....	2
2.4 Document Conventions	2
2.5 Miscellaneous	3
Overall Description	4
2.6 Product Features	4
2.7 Functional description.....	5
2.8 User Classes and Characteristics	6
2.8.1 2.2 User Classes and Characteristics	6
2.9 Design and Implementation Constraints	7
2.10 Assumptions and Dependencies	7
3. Technical Architecture.....	9
3.1 Application and Data Architecture	9
3.2 Component Interactions and Collaborations	14
3.3 Design Reuse and Design Patterns.....	26
3.4 Technology Architecture.....	26
4. Screenshots/Prototype	28
4.1 Workflow	28
4.2 Screens	29
4.3 Additional Information.....	33
5. Other Design Details	33
6. Revised Project Plan	33
7. References.....	34

Revision History

Name	Date	Reason For Changes	Version

Abstract

Vistelligence is an AI-powered web application designed to make the visa application process smoother and more accessible for individuals by analyzing a user's unique profile, including factors like education, financial situation, and career aspirations, to deliver tailored recommendations for visa types, countries, and universities. The platform goes beyond suggestions by predicting visa approval chances, providing interview preparation tools, and offering practical guidance on document preparation, financial planning, and risk management. What sets Vistelligence apart is its focus on enhancing the user experience, with features like a dynamic visa cost estimator to help users budget effectively and an interactive community forum that fosters peer-to-peer support and knowledge sharing, reducing the stress of navigating complex processes while creating a sense of community among users pursuing similar goals. Built on real-time data and advanced AI algorithms, Vistelligence ensures accuracy and reliability in its recommendations, empowering users to make confident decisions about their global mobility journey. Whether seeking to travel, study, or invest abroad, Vistelligence serves as a comprehensive solution, streamlining every step of the process and transforming overwhelming procedures into manageable steps, enabling users to focus on achieving their dreams and building brighter futures on an international stage.

1. Introduction

Vistelligence provides personalized visa type recommendations, country and university suggestions, and visa approval predictions. It includes interview preparation, action plans, and real-time updates on

..

policies. The platform features a user-friendly interface and a community forum for advice-sharing. It simplifies the visa process for students, professionals, and investors. Here's the content for each section regarding the document to explain:

2. Product

Visa applicants often struggle with understanding which visa types suit their needs, predicting approval chances, and preparing the necessary documentation. The existing platforms provide generic advice, leaving applicants without personalized guidance, leading to rejections and inefficiencies. Vistelligence addresses these issues with AI-powered solutions for personalized visa recommendations and end-to-end process support.

2.1 Background

With the increasing demand for international travel, education, and work opportunities, visa application processes have become a crucial aspect of global mobility. As we know that the current tools lack personalization and rely heavily on static information. Integrating artificial intelligence into this domain can transform the experience, reducing uncertainty and improving approval rates.

2.2 Objective(s)/Aim(s)/Target(s)

The objectives of this platform are:

- It simplifies visa application processes using AI.
- It provides personalized, actionable recommendations for visas, countries, and universities.
- It enhances the success rate of applications with data-driven insights.
- It provides a community for peer-to-peer learning and expert support.

2.3 Scope

Vistelligence covers the entire visa application life-cycle:

- Personalized visa, country, and university suggestions.
- AI-based approval predictions and interview preparation.
- Comprehensive application strategies and real-time cost estimations.
- Community forums for shared experiences and collaborative learning.

This platform caters to students, professionals, and investors seeking visas worldwide. Vistelligence offers an AI-driven platform that simplifies the visa application process, providing personalized recommendations for visa types, universities, and countries. It builds trust by focusing on high success rates and personalized strategies for global mobility.

2.4 Document Conventions

This Software Requirements Specification (SRS) follows the typographical and formatting conventions outlined below for clarity and consistency:

- **Font Style:** The font used throughout the document is Times New Roman.
- **Font Size:**
 - The **body text** is set at **12-point size** to ensure readability.
 - **Subheadings** are styled with a **16-point size** for better differentiation and hierarchy.
 - **Major page titles** or **section headers** are formatted with a **28-point size** to highlight important sections and maintain a clear structure.
- **Bold Text:** Bold font is used to highlight **key terms** or **important sections** for quick reference.
- **Bullet Points and Numbering:** Bullets and numbered lists are used for clarity and ease of reference in sectioned or grouped information.
- In **Reference section** the name of articles or books are in italic.
- **Classes** names are also in italic.

2.5 Miscellaneous

Team Communication:

Weekly Zoom meetings and a dedicated Whats App group will be used for updates and discussions to maintain coordination.

Version Control:

GitHub will be utilized for collaborative development and version management, ensuring seamless teamwork.

Documentation Standards:

All documentation will follow IEEE standards to maintain clarity and consistency throughout the project.

Design Tools:

Canva will be used for UI/UX mockups, and Draw.io for creating diagrams and visual models.

Overall Description

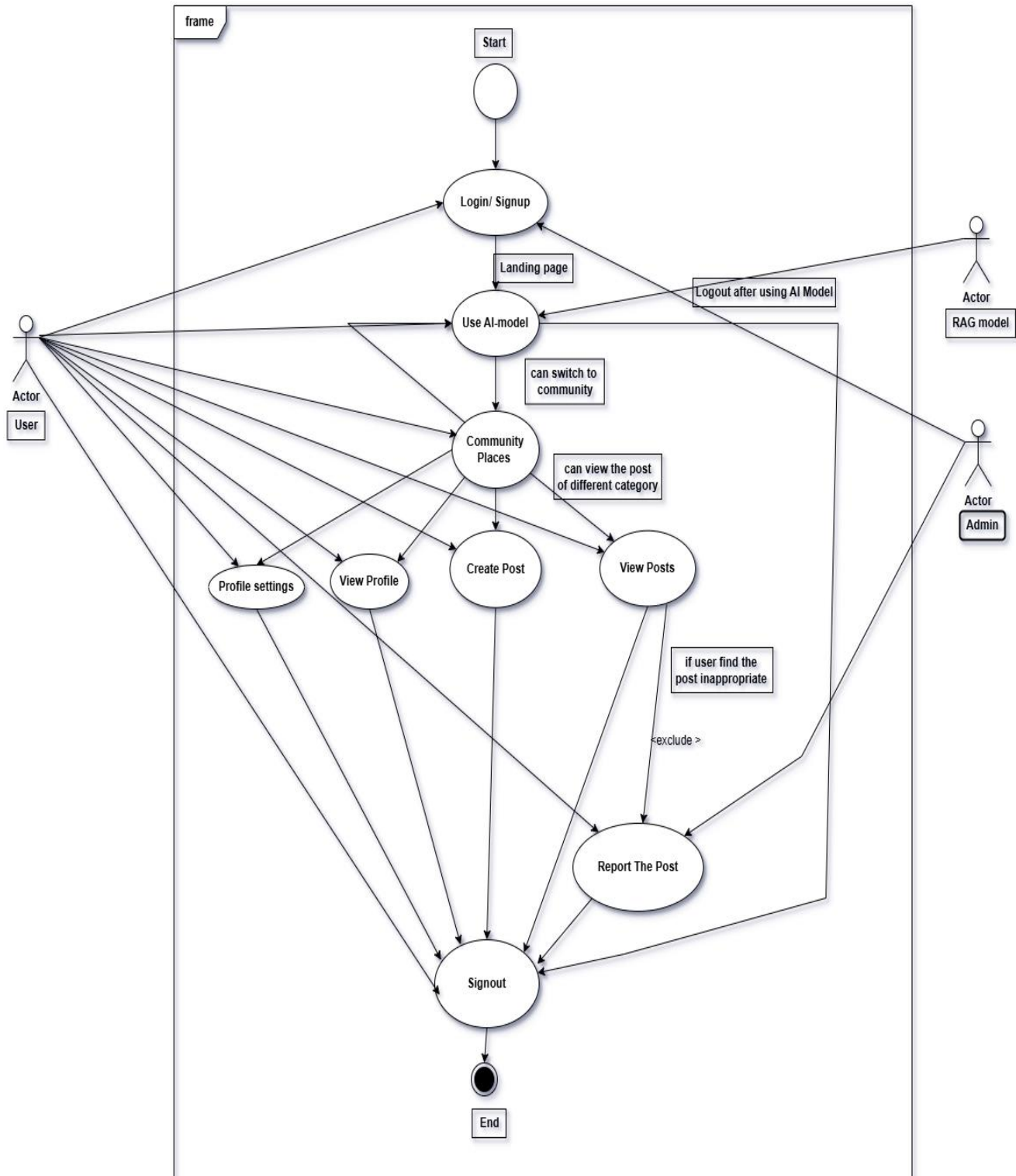
Overall description of “Vistelligence” explained below:

2.6 Product Features

Vistelligence offers several key features to help users with their visa application process. These include:

- Personalized visa type recommendations based on user information.
- Suggestions for the best countries and universities for studying or working.
- Visa approval chance predictions to help users plan ahead.
- Interview preparation tips and strategies.
- Personalized action plans for successful applications.
- A community forum for users to connect and share advice.
- Real-time updates on visa policies and regulations.
- Simple, user-friendly interface with clear guidance at every step.

2.7 Functional description



2.8 User Classes and Characteristics

Vistelligence is designed for multiple user classes, each with unique needs and responsibilities

2.8.1 2.2 User Classes and Characteristics

Vistelligence is designed for multiple user classes, each with unique needs and responsibilities. THE breakdown of each user clas is given below:

1. *Students*

Characteristics:

- Primarily focused on obtaining visa information for study purposes.
- Need guidance on visa types, countries, and universities that best suit their academic goals.
- Typically have lower technical expertise and need a simple, user-friendly interface.
- Require personalized recommendations and visa approval predictions.

Responsibilities:

- Use the app to explore visa options, universities, and countries.
- Track visa approval chances and access resources for interview preparation.
- Create and manage their user profiles for tailored suggestions.

2. *Professionals*

Characteristics:

- Interested in obtaining visa information related to employment and career development.
- Likely to have a higher level of technical expertise compared to students.
- Need more detailed information about work visas, approval predictions, and interview strategies.
- May have stricter privacy and security requirements due to sensitive career-related data.

Responsibilities:

- Use the app to explore visa options for employment purposes.
- Monitor visa approval chances and access resources for interview prep.
- Maintain privacy and security of their application data and personal information.

3. *General Public*

Characteristics:

- Casual users who want general visa information or are exploring visa options.
- May not require detailed or personalized information but still benefit from basic suggestions and guidance.
- Generally, have basic technical expertise and require a simple interface.

Responsibilities:

- Access general visa information and stay updated on visa regulations.
- Use the basic features of the app to explore visa types and countries.
- Benefit from updates on visa rules and real-time policy changes.

4. Admin

Characteristics:

- Responsible for managing and overseeing the operation of the app.
- Typically have the highest level of access and technical expertise.
- Ensures that the platform runs smoothly, with no issues related to user data or system performance.
- Required to monitor, moderate, and update content related to visa policies, application strategies, and the community forum.

Responsibilities:

- **User Management:** Admins will have the ability to manage and oversee user profiles for all classes, ensuring data integrity and security.
- **Content Moderation:** Monitor and update content in the community forum, ensuring that all information provided is accurate and relevant.
- **System Management:** Responsible for system performance and troubleshooting. They ensure that the platform runs efficiently and securely.
- **Access Control:** Set and manage different levels of access for users, ensuring the right privileges for each class (e.g., students, professionals, immigration advisors).
- **Real-time Updates:** Manage updates regarding visa policies, regulations, and any external data sources the app relies on.
- **Reporting and Analytic:** Admins will have access to user activity and app performance analytic to improve functionality and user experience.
- **Security Management:** Ensure that all user data is securely stored and managed, complying with relevant regulations like GDPR.

2.9 Design and Implementation Constraints

The development of Vistelligence will need to consider:

- **Regulatory Policies:** The app must comply with data protection regulations (e.g., GDPR) to protect user privacy.
- **Platform Compatibility:** The app needs to work on multiple devices and browsers with no compatibility issues.
- **Security:** Strong security measures to protect user data and ensure the safety of personal information.
- **Technology:** The app will be built using modern technologies like HTML, CSS, JavaScript (for front-end), and Node.js as the server-side environment, with Express.js for building Restful APIs (for back-end). Mongo DB will be used as the database.
- **User Experience:** The app must be easy to navigate and offer a smooth user experience for all user classes, from students to professionals.

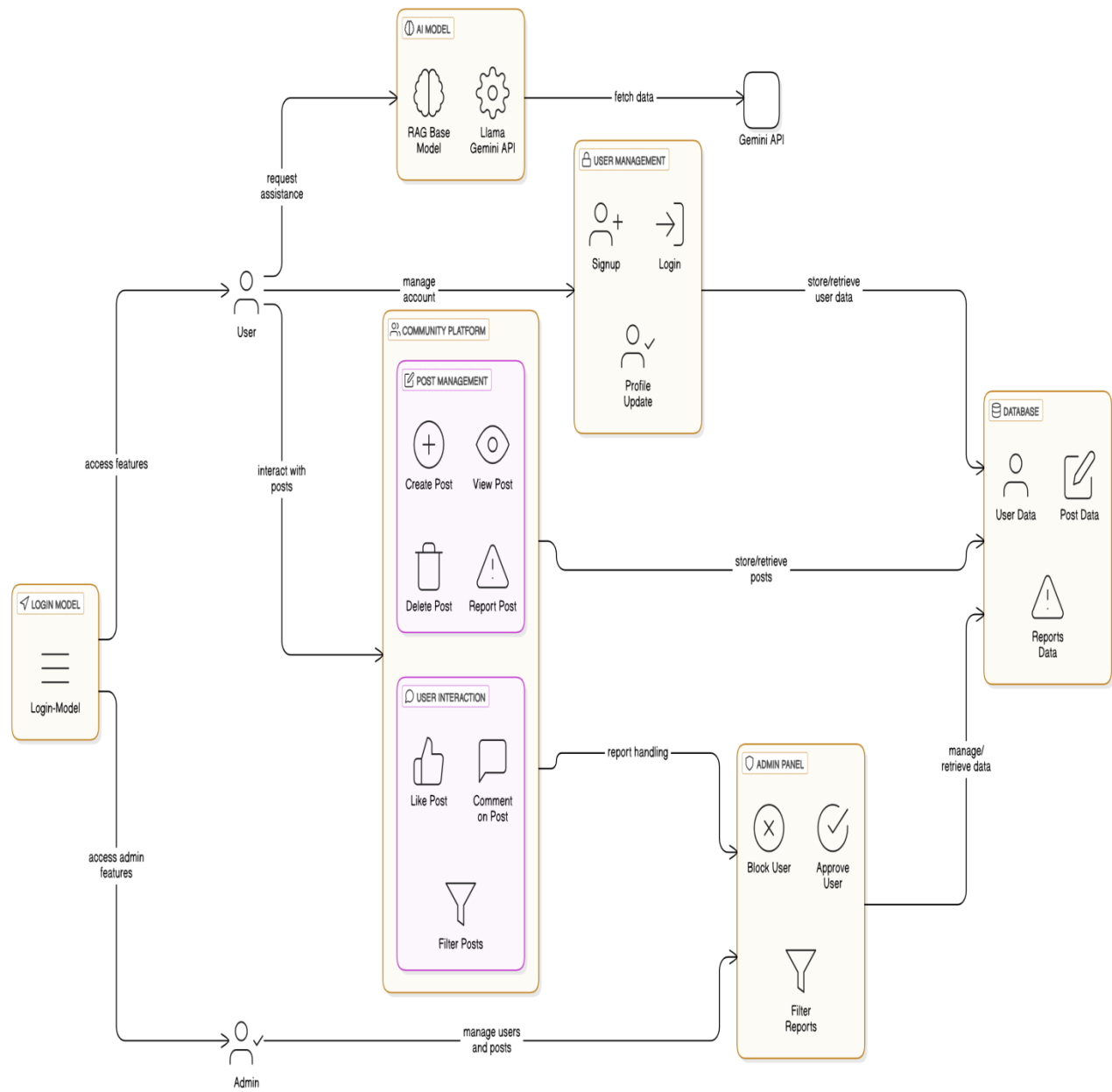
2.10 Assumptions and Dependencies

The development of Vistelligence will need to consider some assumptions and dependencies:

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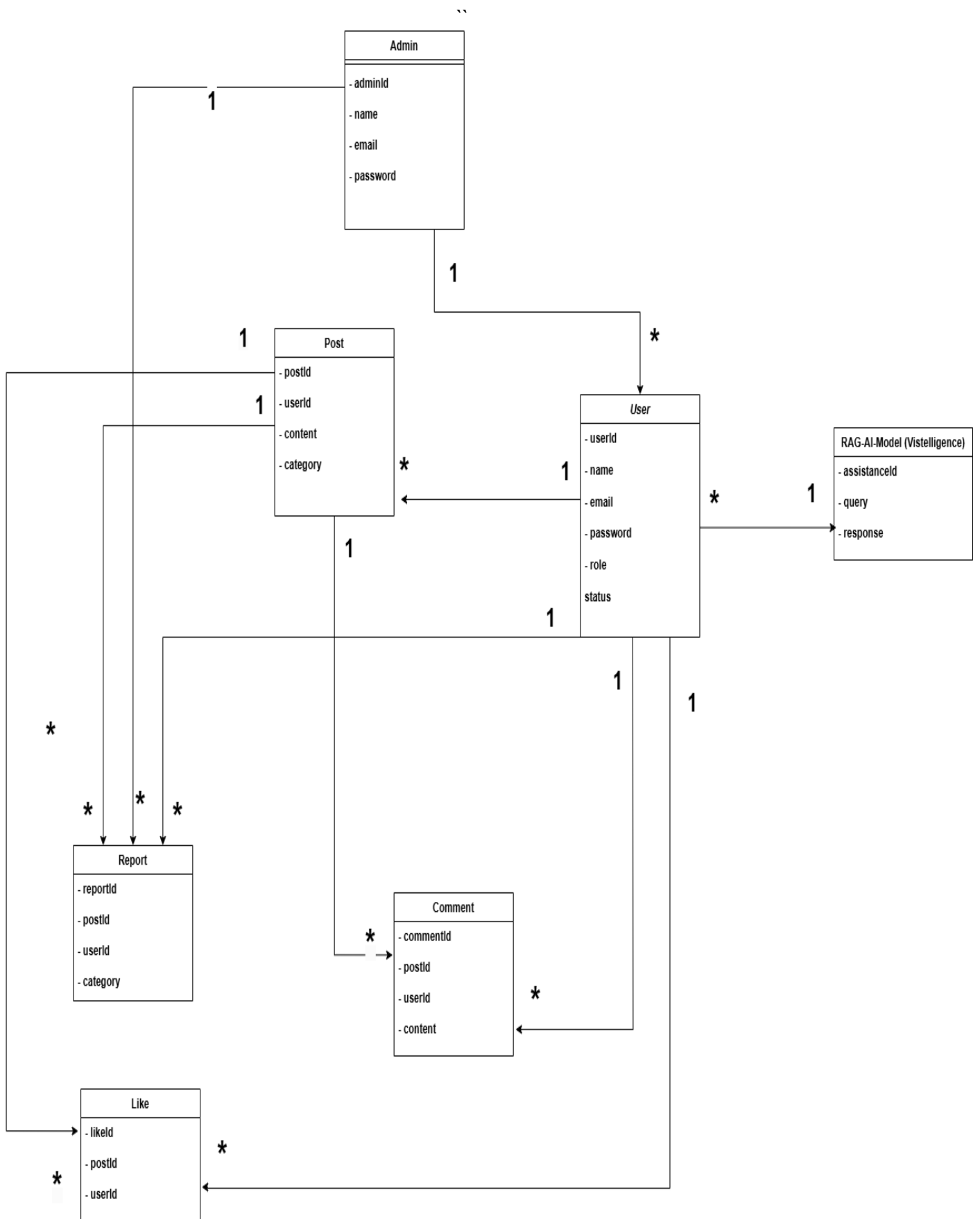
- **Third-party APIs:** The app will rely on APIs to get real-time visa policy updates and country-specific data.
- **User Behavior:** It's assumed that users will have internet access to use the app's full features.
- **External Components:** The app may depend on external services for specific visa information or government regulations that are not available directly through the app.
- **Device Requirements:** The app assumes that users will have modern devices with basic capabilities like internet browsing and data entry

3. Technical Architecture

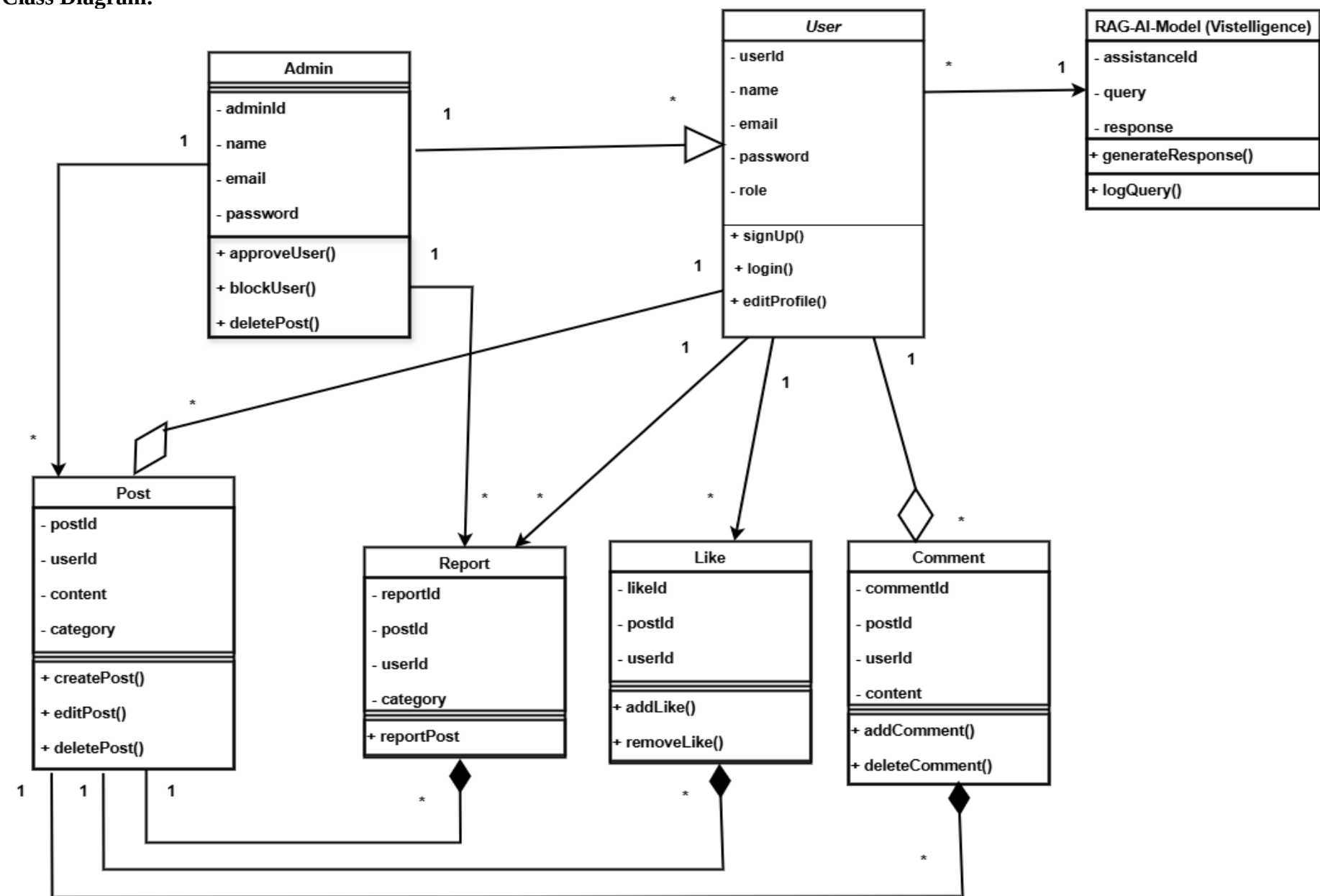


3.1 Application and Data Architecture

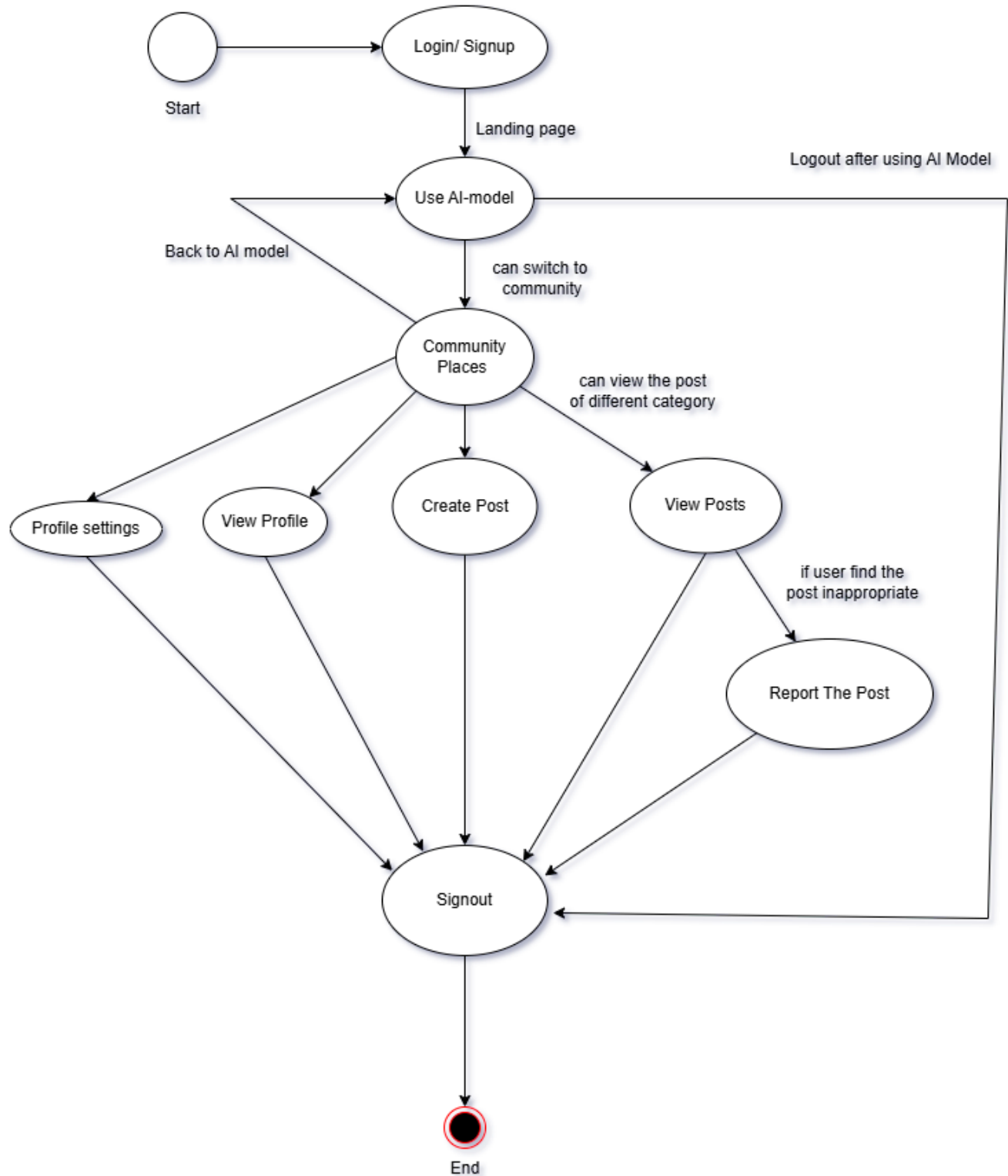
ERD diagram:



Class Diagram:



Activity Diagram:



Decision Tables:

<F24SE055>

Login/Signup Process:

Condition	Is Email Valid?	Is Password Strong?	Is Username Unique?	Action
Case 1	Yes	Yes	Yes	Register user
Case 2	Yes	Yes	No	Prompt "Username already taken"
Case 3	Yes	No	Yes	Prompt "Weak password"
Case 4	Yes	No	No	Prompt "Weak password and username taken"
Case 5	No	Any	Any	Prompt "Invalid email"

AI Model Usage

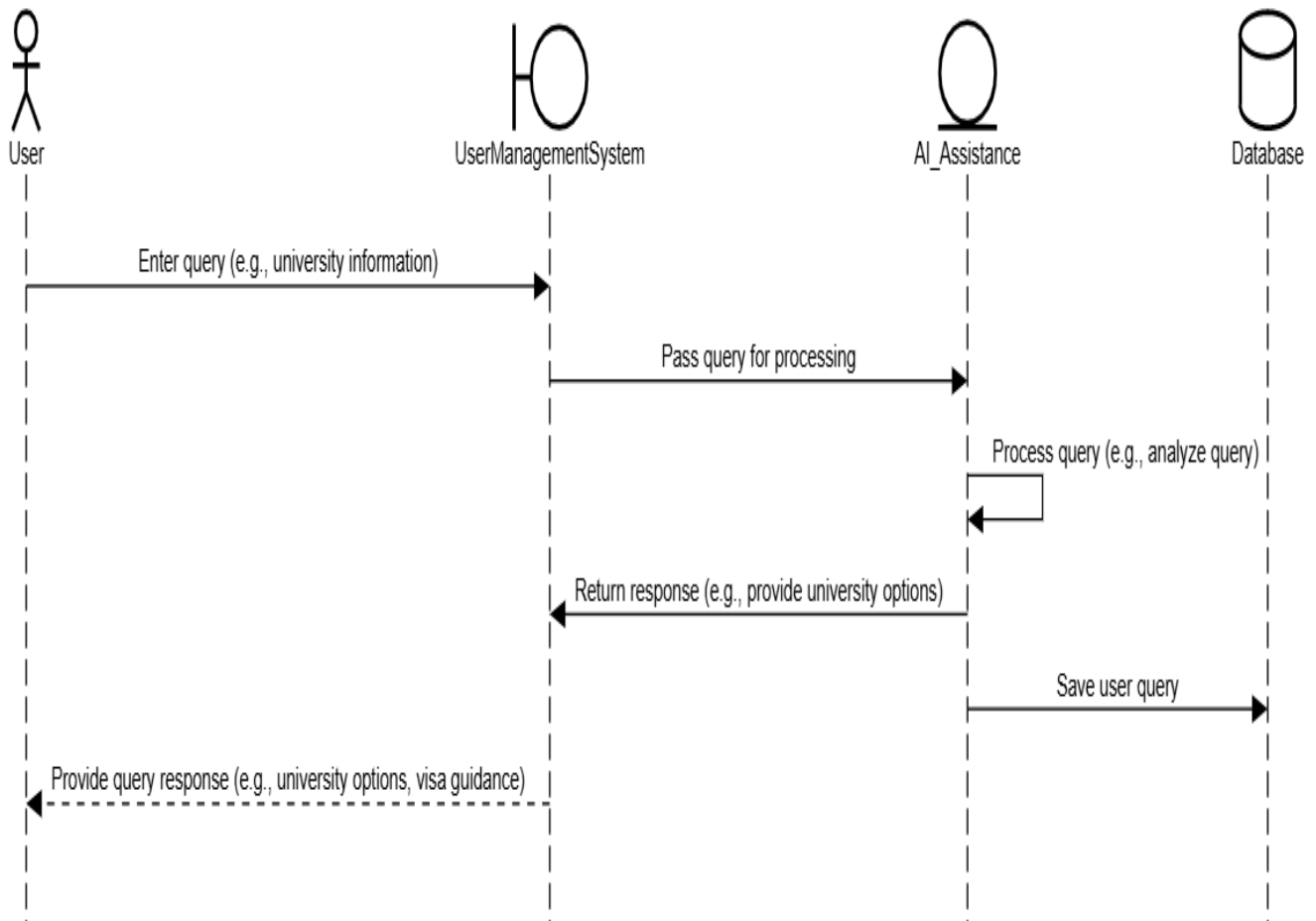
Condition	User Logged In?	Model Available?	User Request Valid?	Action
Case 1	Yes	Yes	Yes	Grant access to AI model
Case 2	Yes	Yes	No	Prompt "Invalid request"
Case 3	Yes	No	Any	Prompt "Model currently unavailable"
Case 4	No	Any	Any	Redirect to login/signup

Community Places:

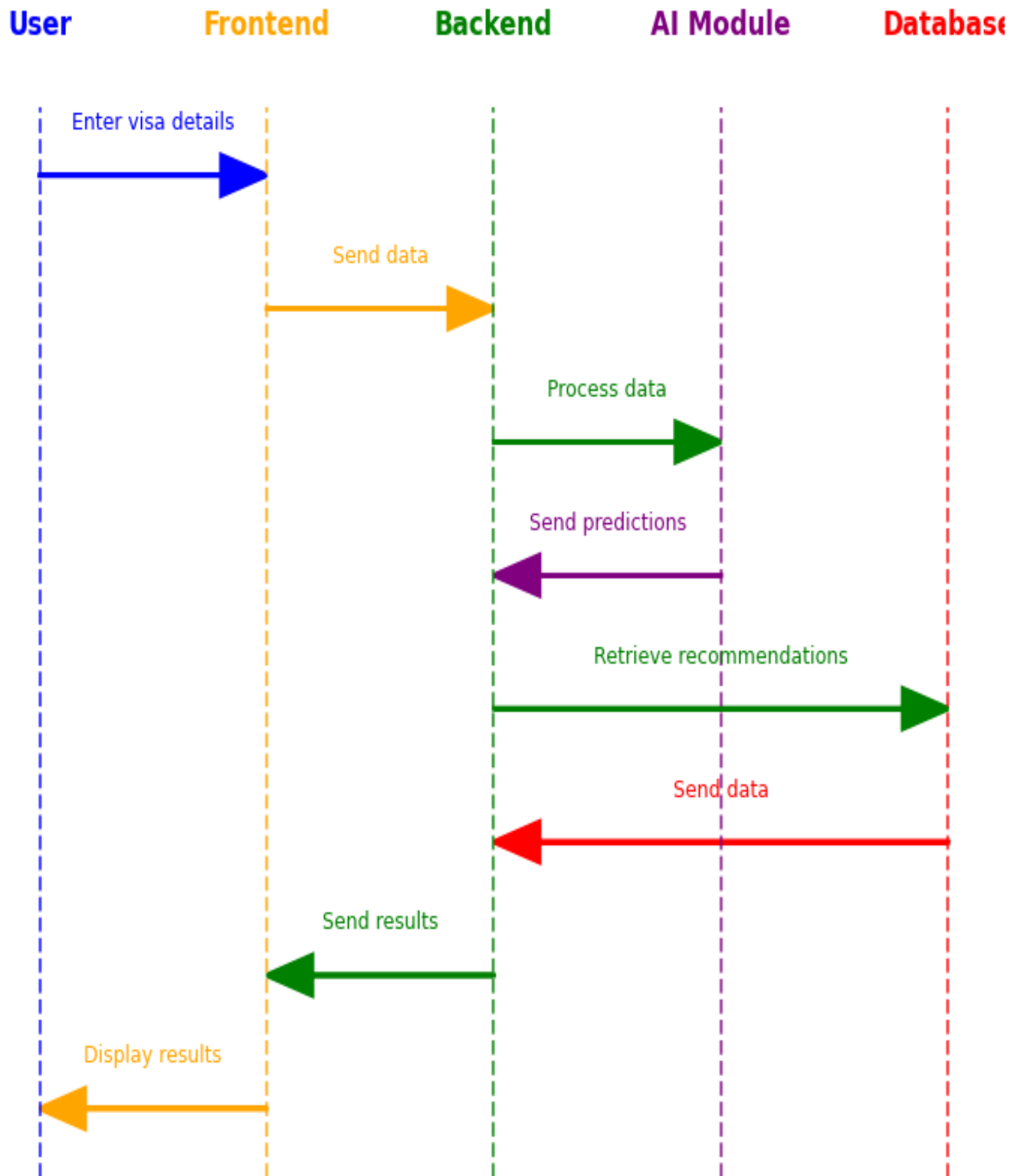
Condition	Post Valid?	Category Available?	User Post?	Report	Action
Case 1	Yes	Yes	No		Display posts
Case 2	Yes	Yes	Yes		Mark post as inappropriate
Case 3	No	Any	Any		Prompt "Invalid post content"
Case 4	Any	No	Any		Prompt "Category unavailable"

3.2 Component Interactions and Collaborations

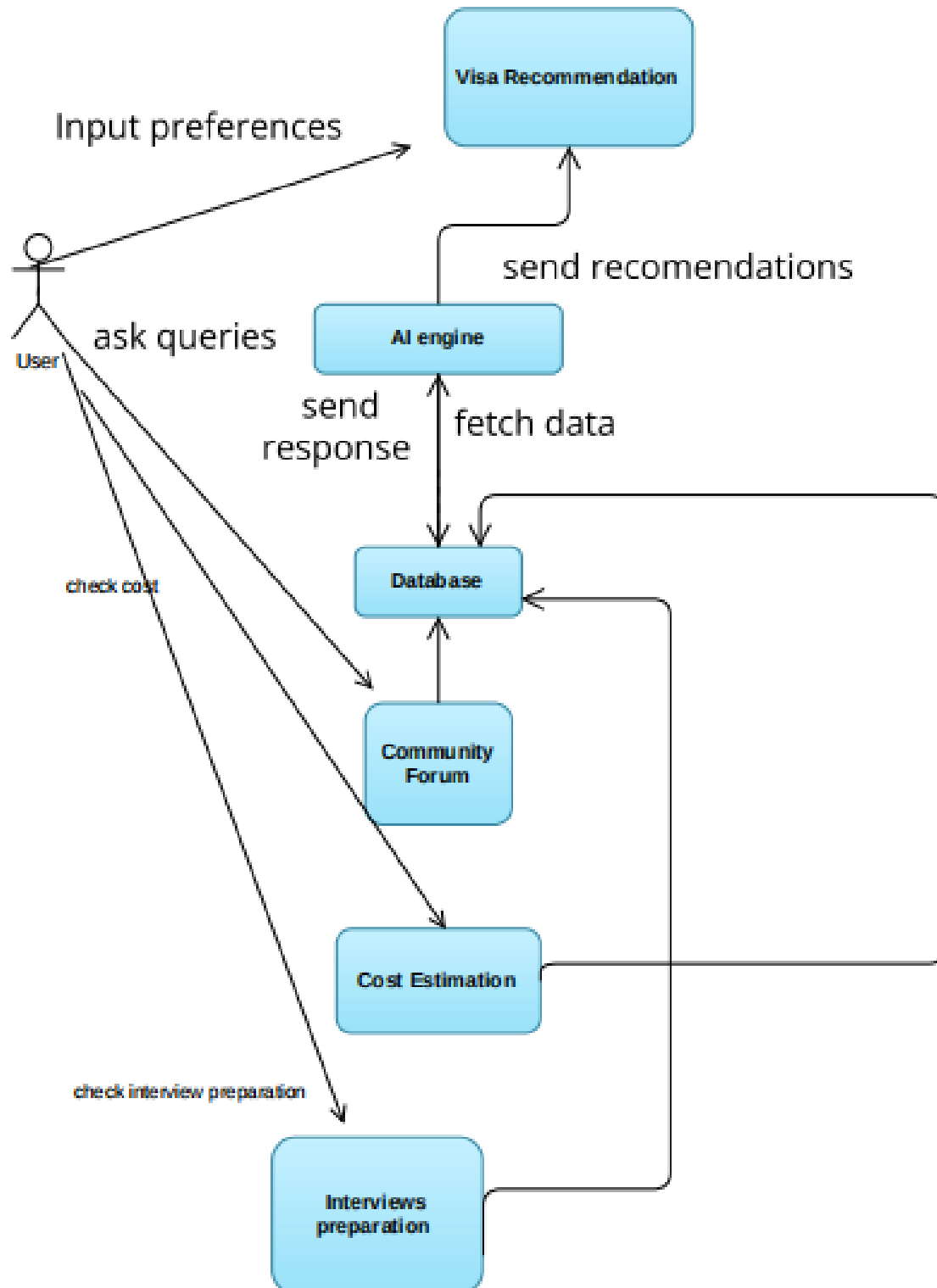
AI interaction model:



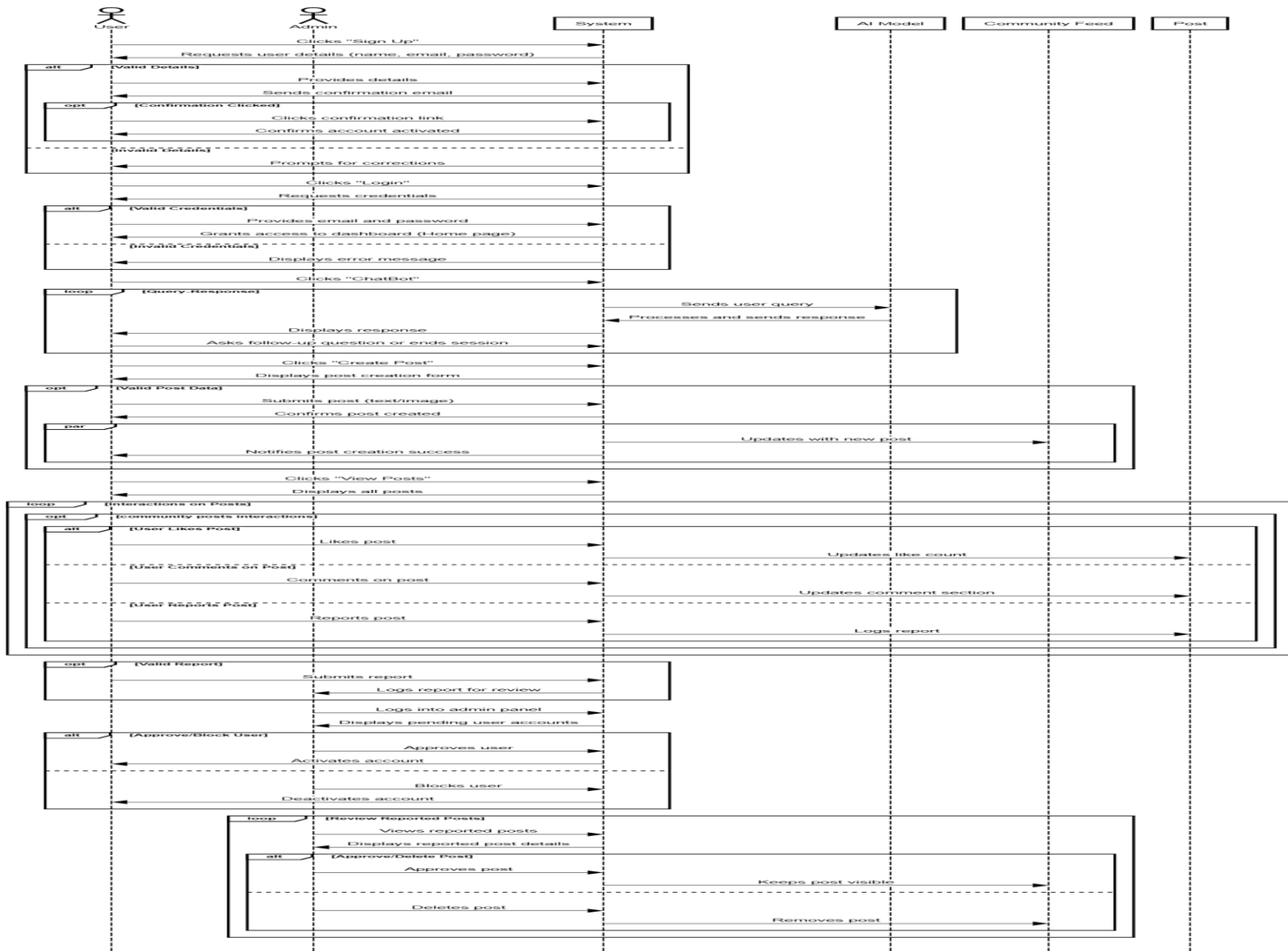
Event trace diagram:

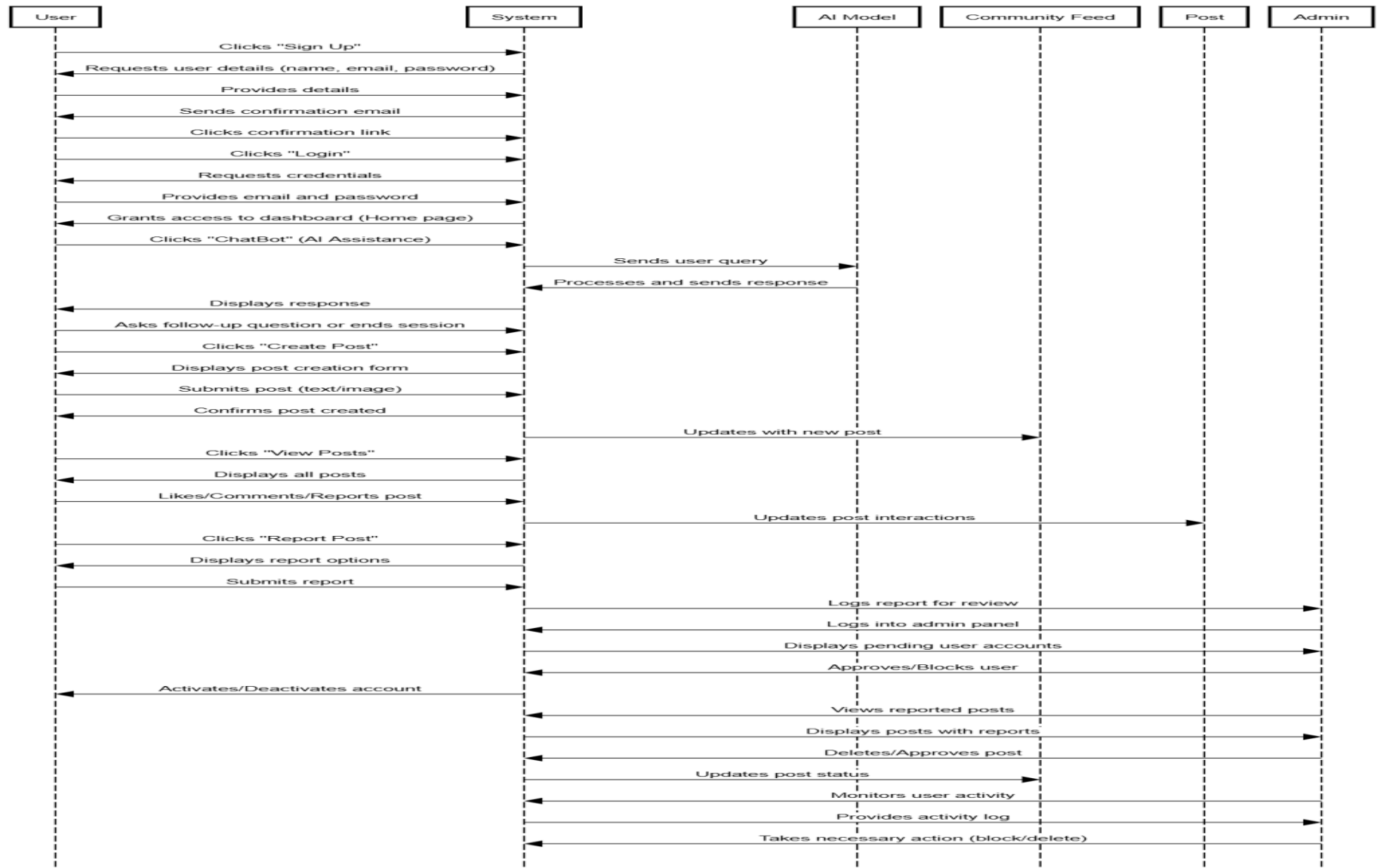


Collaboration diagram:

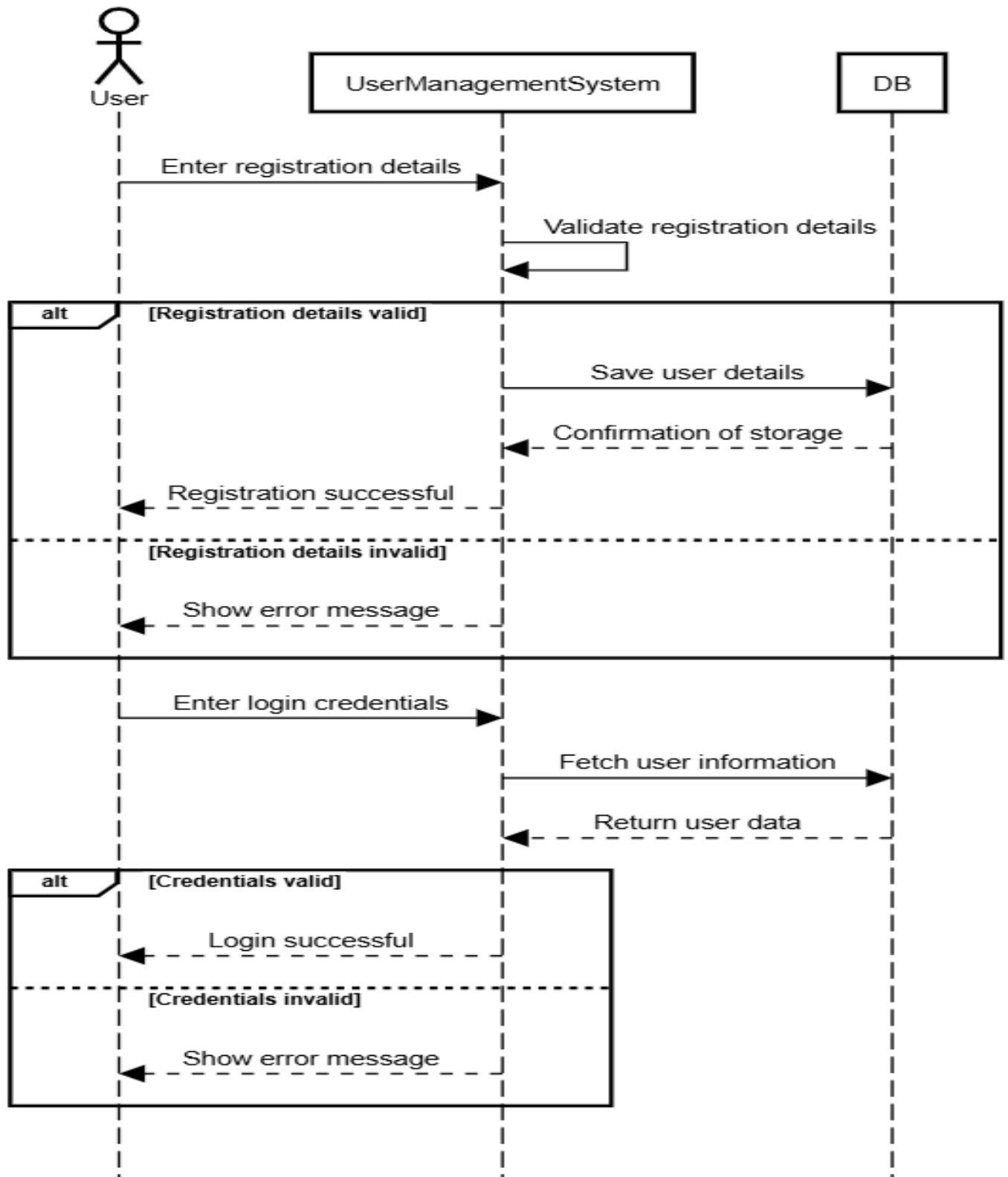


Sequence diagram

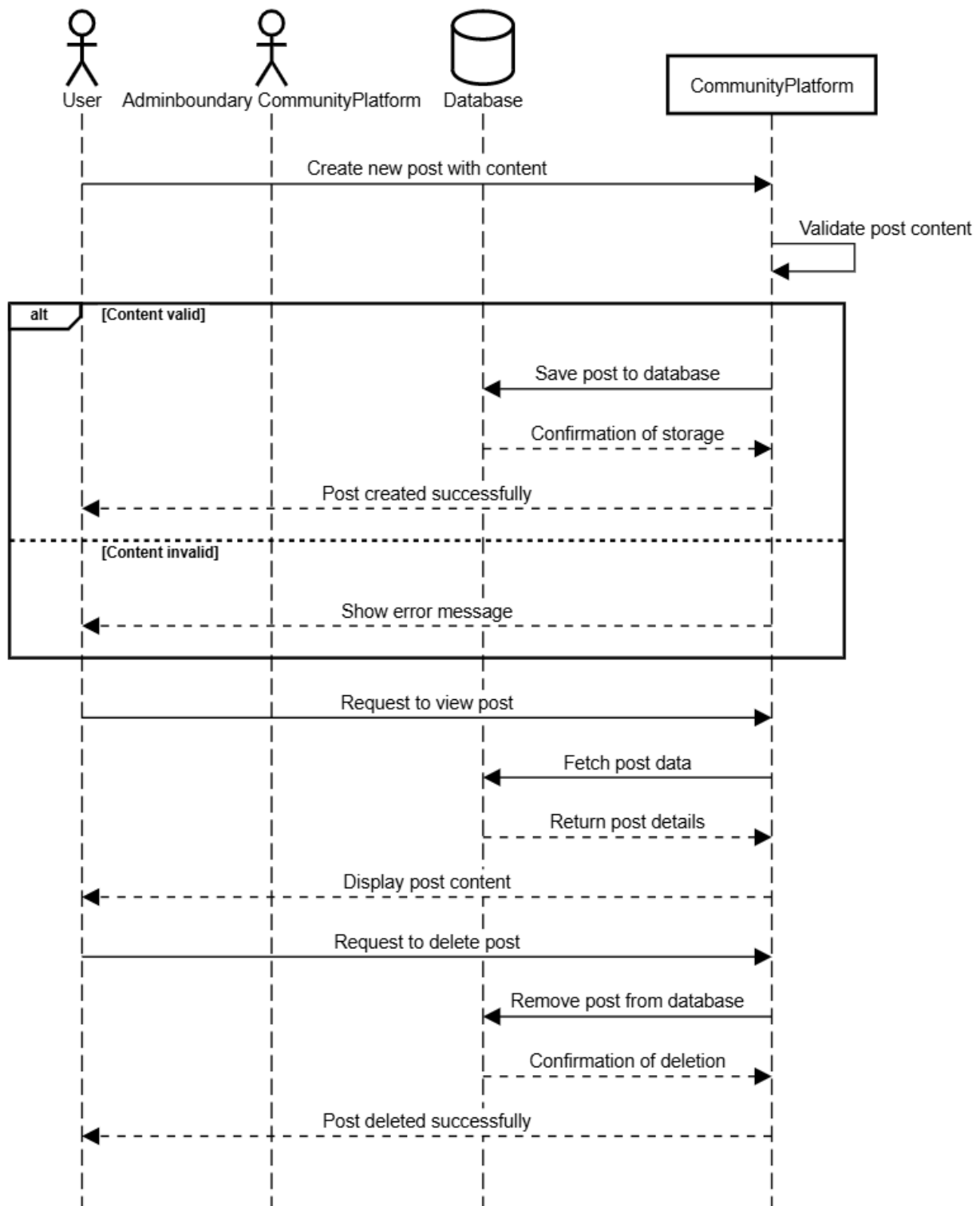




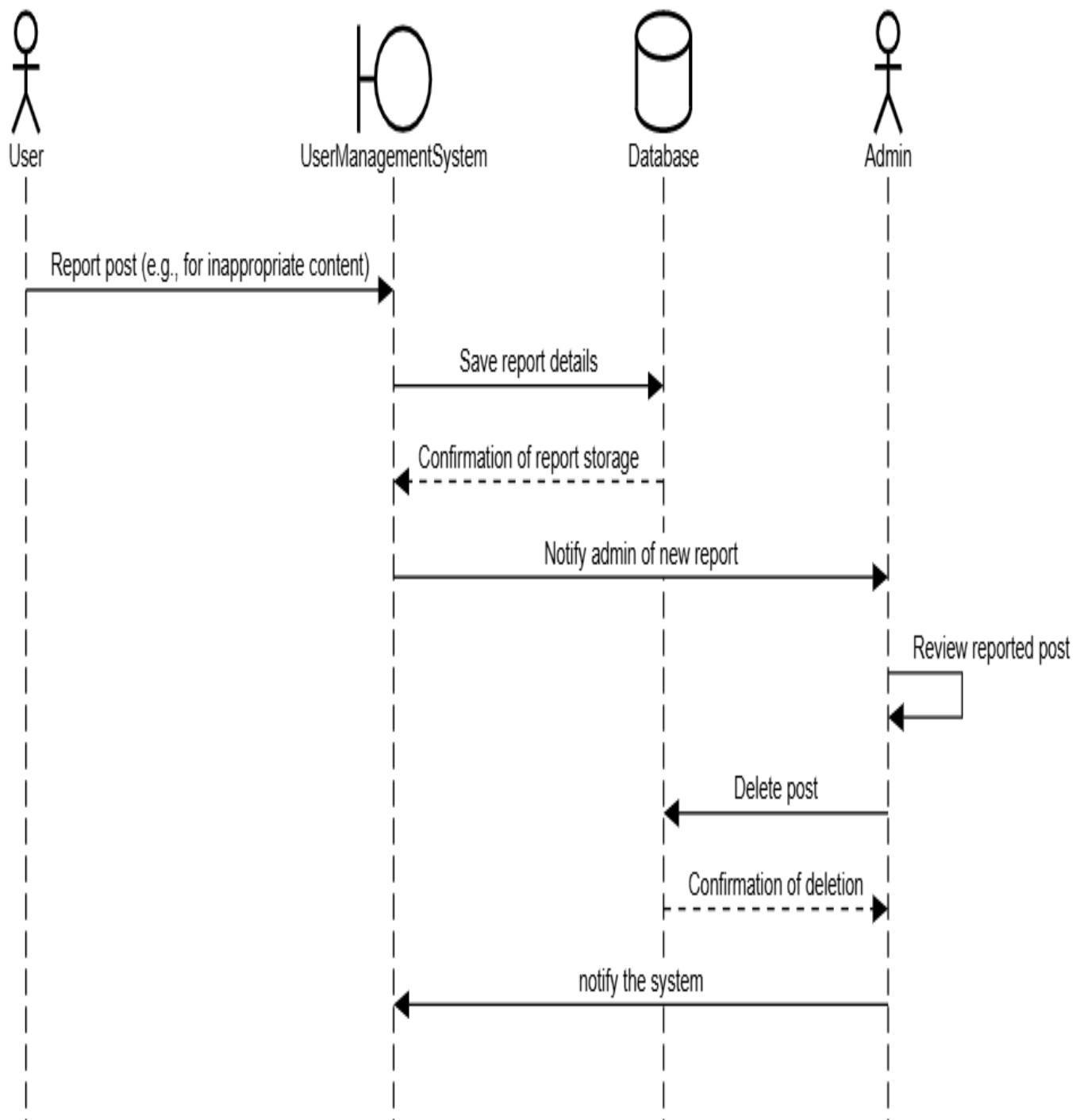
User Management



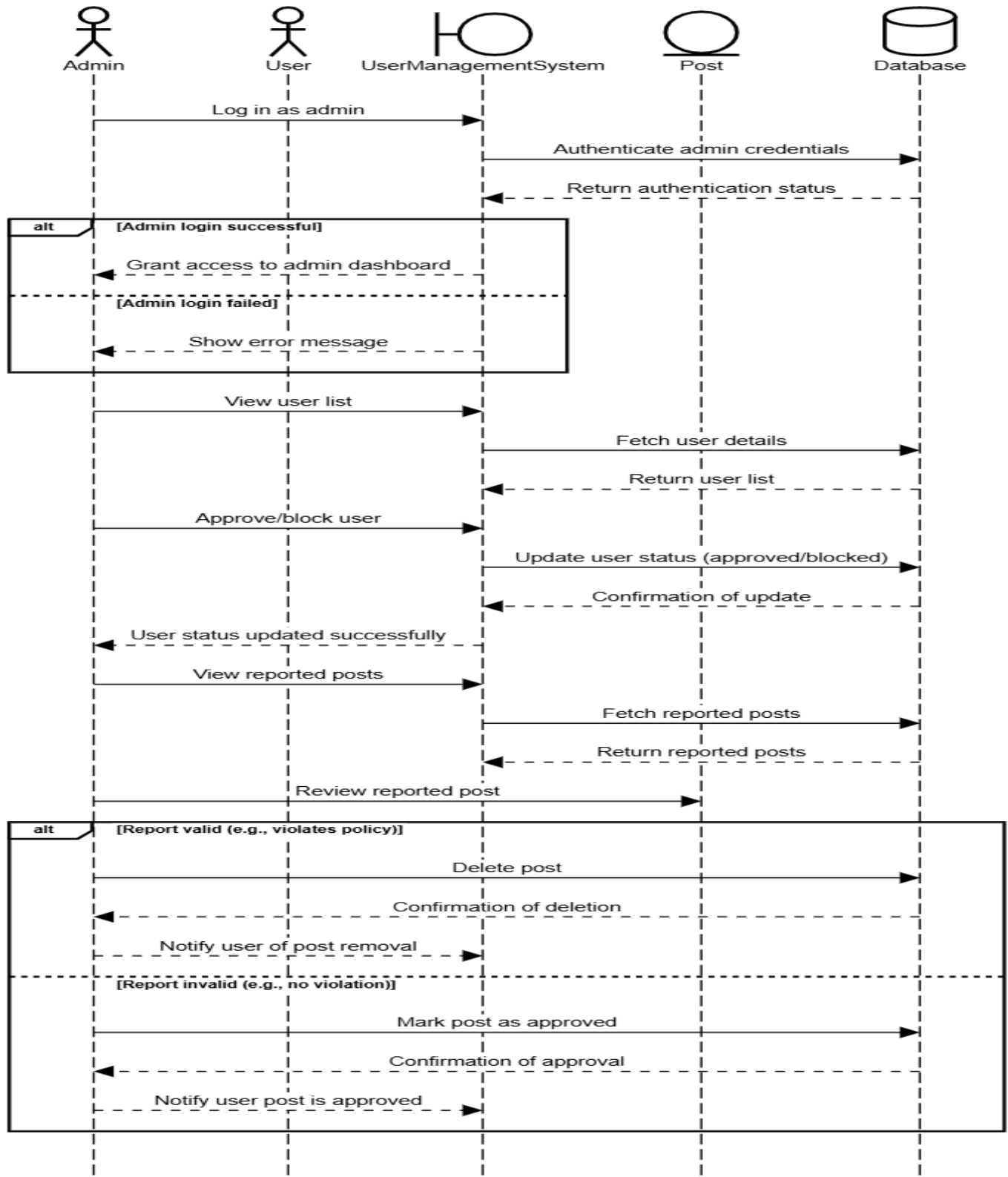
Sequence diagram: Creating and Managing Posts



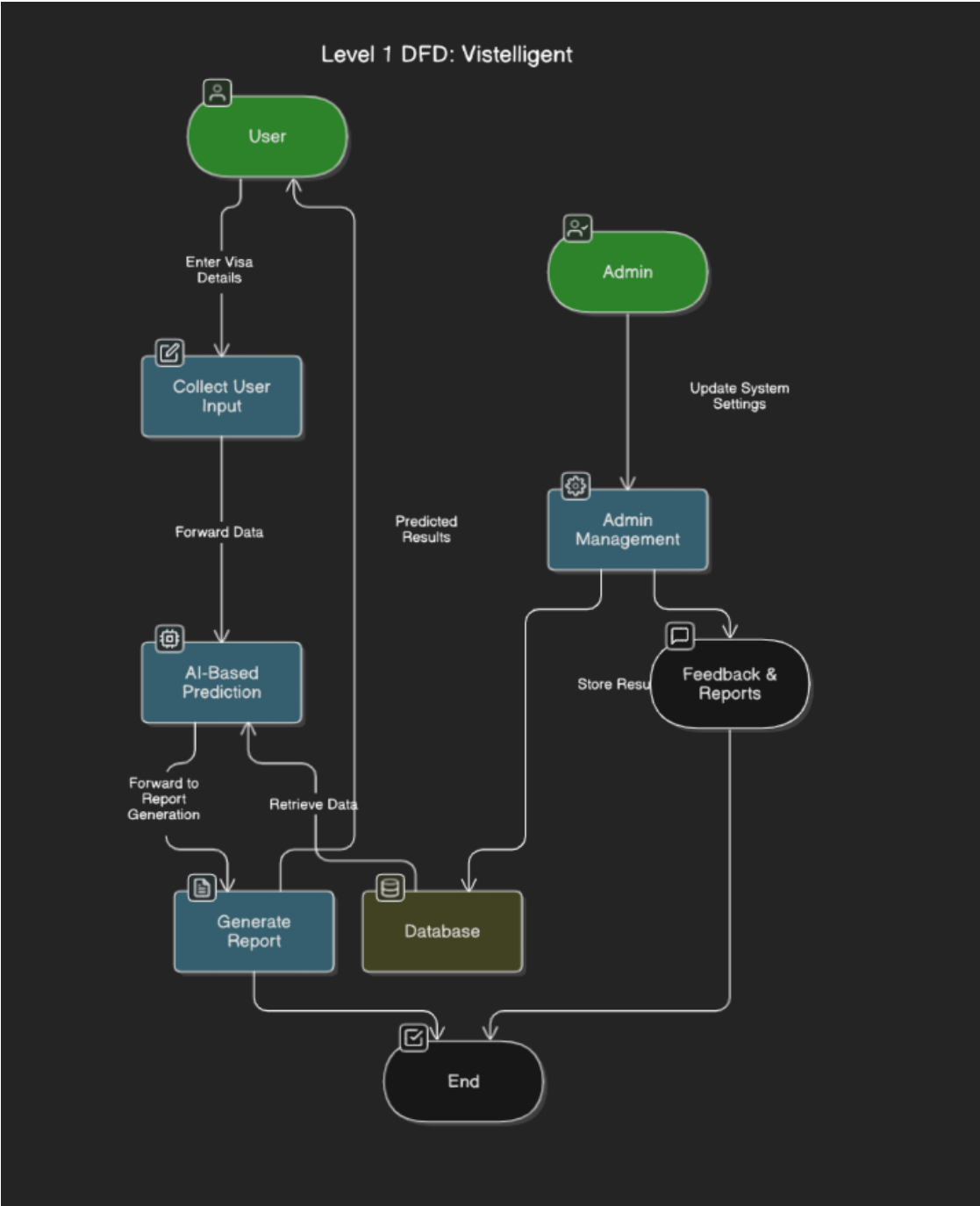
Sequence diagram: report the post



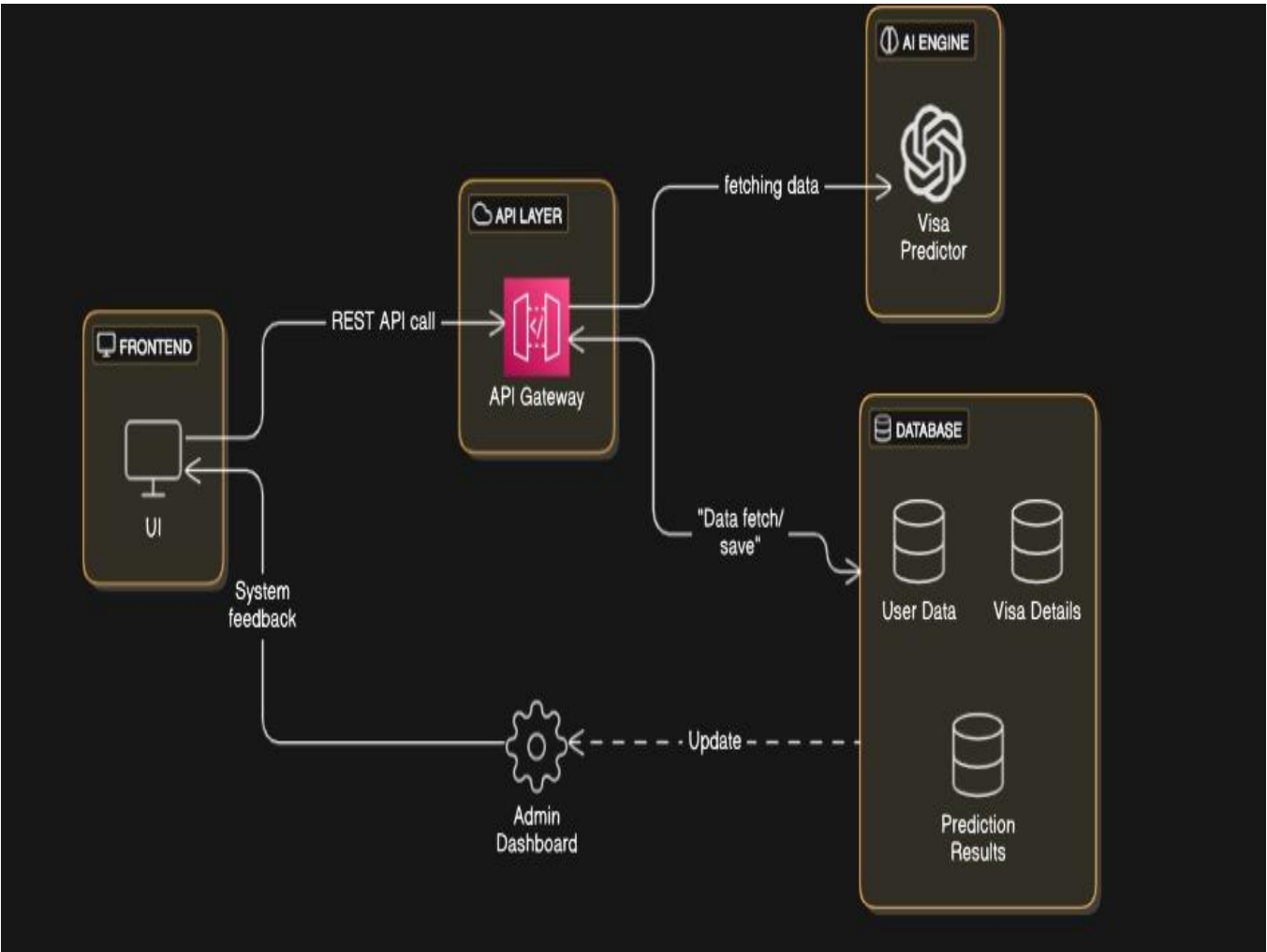
Sequence diagram: admin post review



DataFlow diagram level 1



Component diagram:



3.3 Design Reuse and Design Patterns

Reusable Components:

- **Authentication Module:** A pre-exist, modular authentication system is reused to handle user login, registration, and session management.
- **Payment Gateway Integration:** Reuse of a standard API integration for payment processing, ensuring compatibility and reliability.

Design Patterns:

- **Singleton Pattern:** Applied in managing configuration settings and logging services to ensure a single, globally accessible instance.
- **Factory Pattern:** Used for creating objects.
- **Observer Pattern:** Implemented for real-time updates, such as notifying users about their queries and suggestions (community forum)

User Interface Templates:

- Standardized UI components (buttons, forms, and dashboards) from a library like Bootstrap were reused to ensure a consistent and responsive design.

Database Schemas:

- Predefined schema templates were adapted for managing user data, portfolios, and transactions, reducing the time required for database design.

3.4 Technology Architecture

Platform

- **Front-end:** Built using HTML, CSS, and JavaScript frameworks like React for a responsive and user-friendly interface.
- **Backend:** Powered by Node.js to handle server-side tasks and manage APIs.

System Hosting

- Hosted on cloud platforms like AWS or Google Cloud for reliability and scalability.
- Databases such as MySQL or MongoDB store user and transaction data.

Connectivity Requirements

- Requires stable internet for user interactions and data syncing.
- Uses HTTPS for secure communication and integrates with third-party APIs like payment processors.

Modes of Operation

Online Mode:

- Two-factor authentication (2FA) and encryption protect user data.
- Regular backups and recovery plans ensure safety.

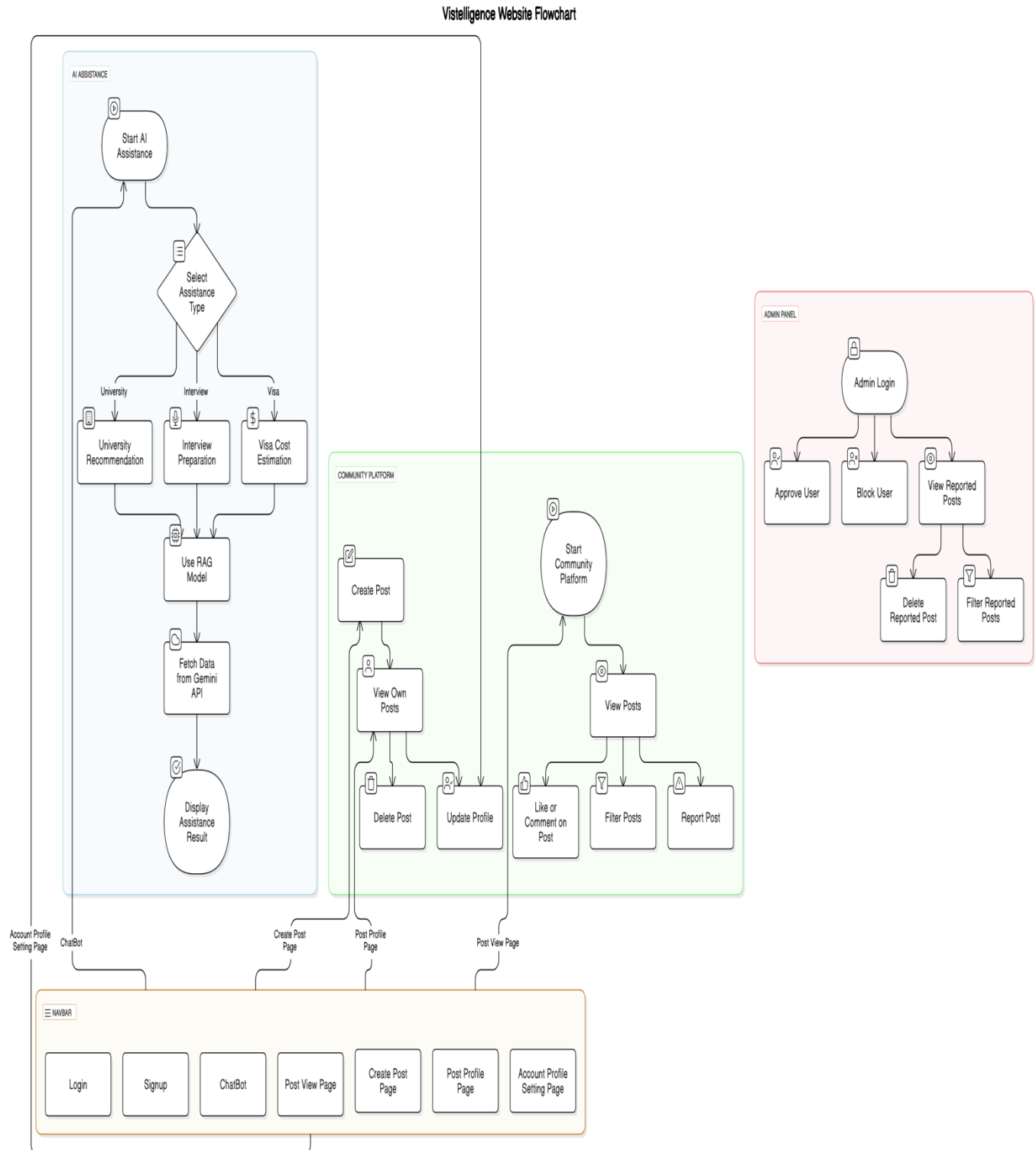
Network

- Globally accessible over the internet with secure HTTPS connections.

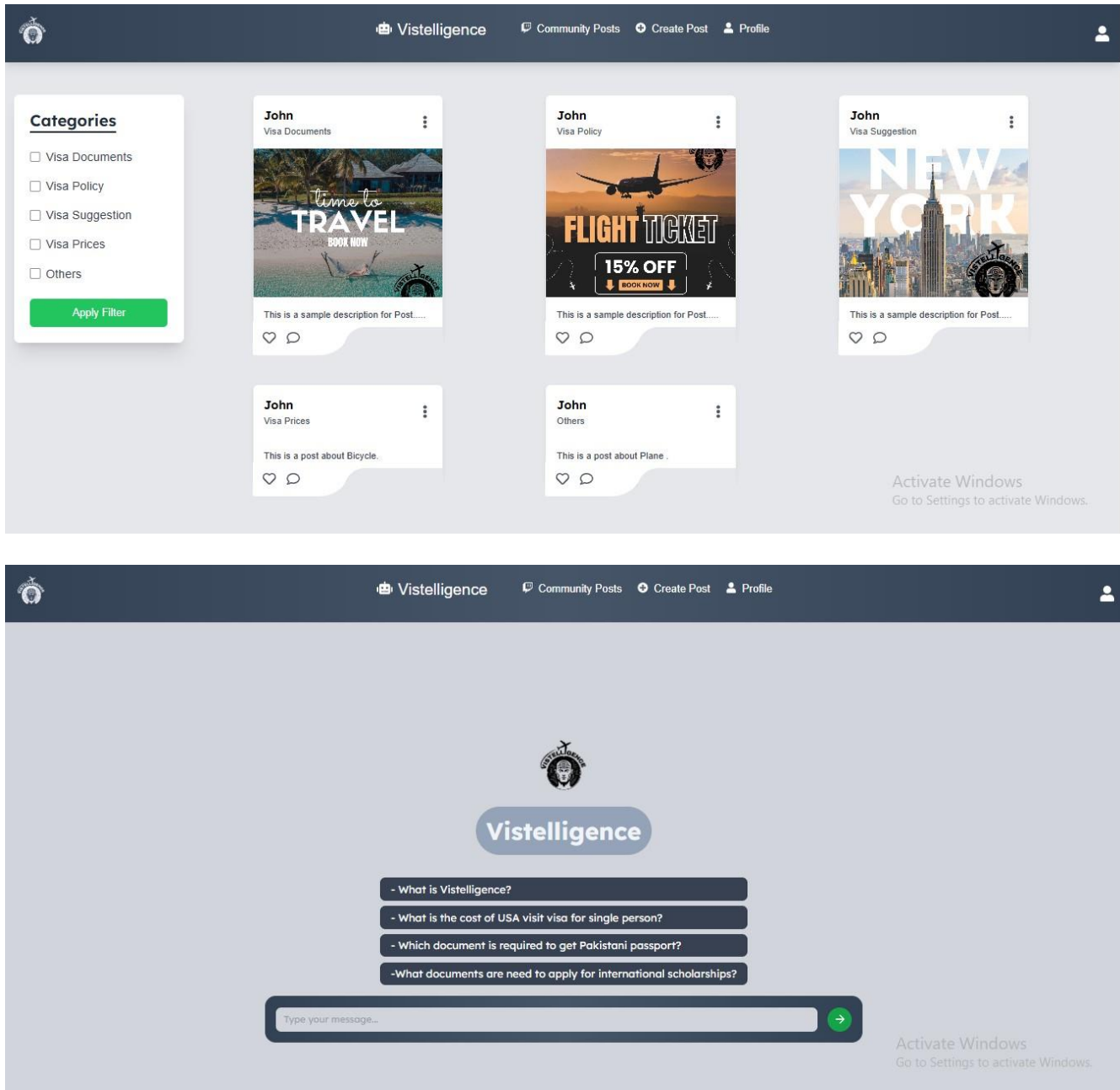
4. Screenshots/Prototype

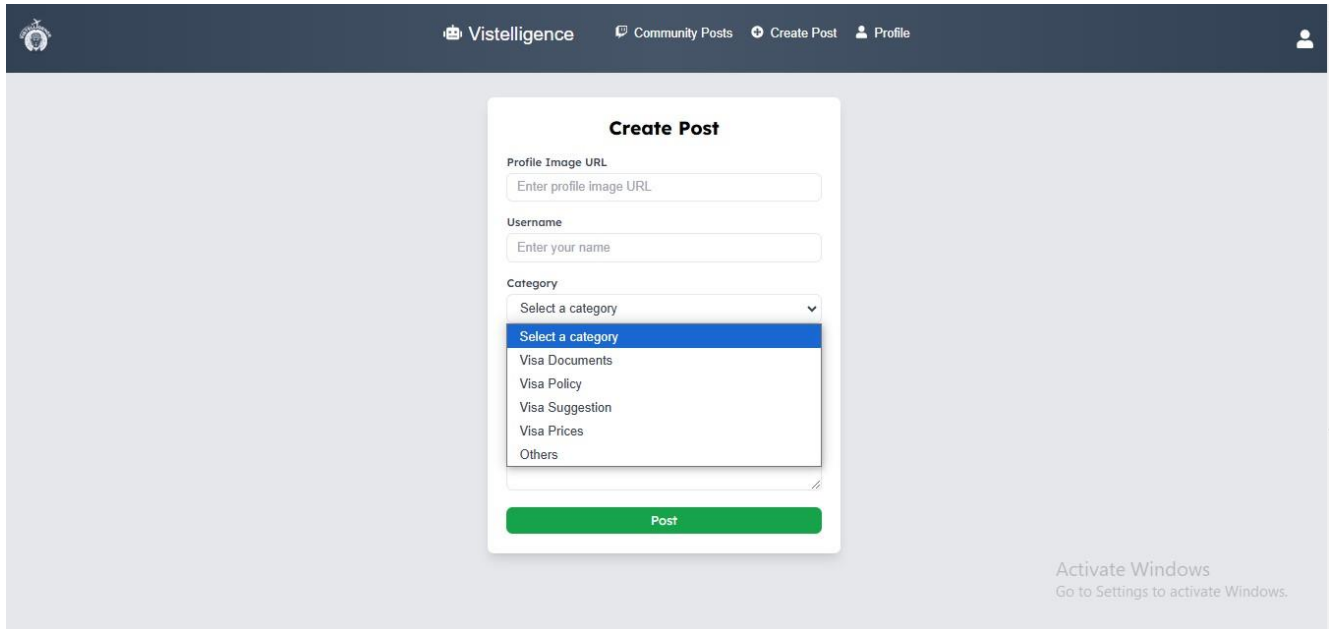
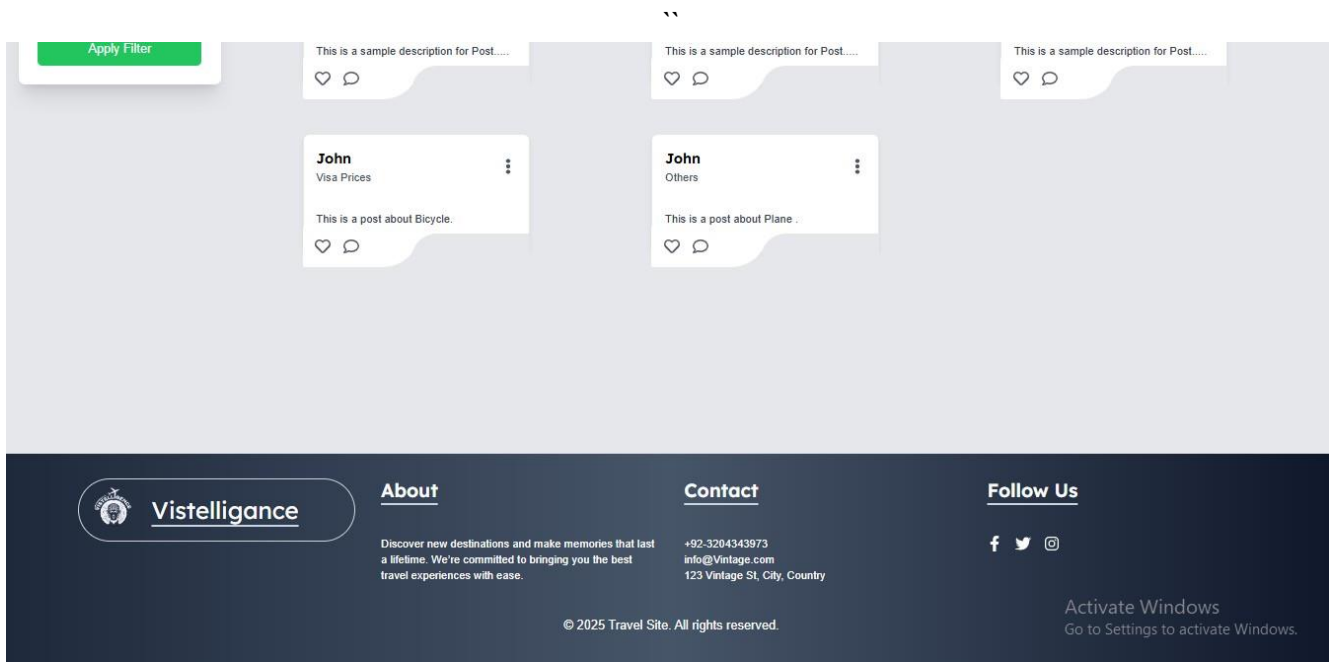
4.1 Workflow

Swimlane diagram:



4.2 Screens







VistelligenceCommunity PostsCreate PostProfile



John

Posts



Booking the Flight Ticket.....

The visa booking visit our site.



Booking The Ticket To Travel.

The visa booking visit our site.



Booking The New York Visa.

Activate Windows
Go to Settings to activate Windows.



Setting

Name

Abdul Wahab

Email

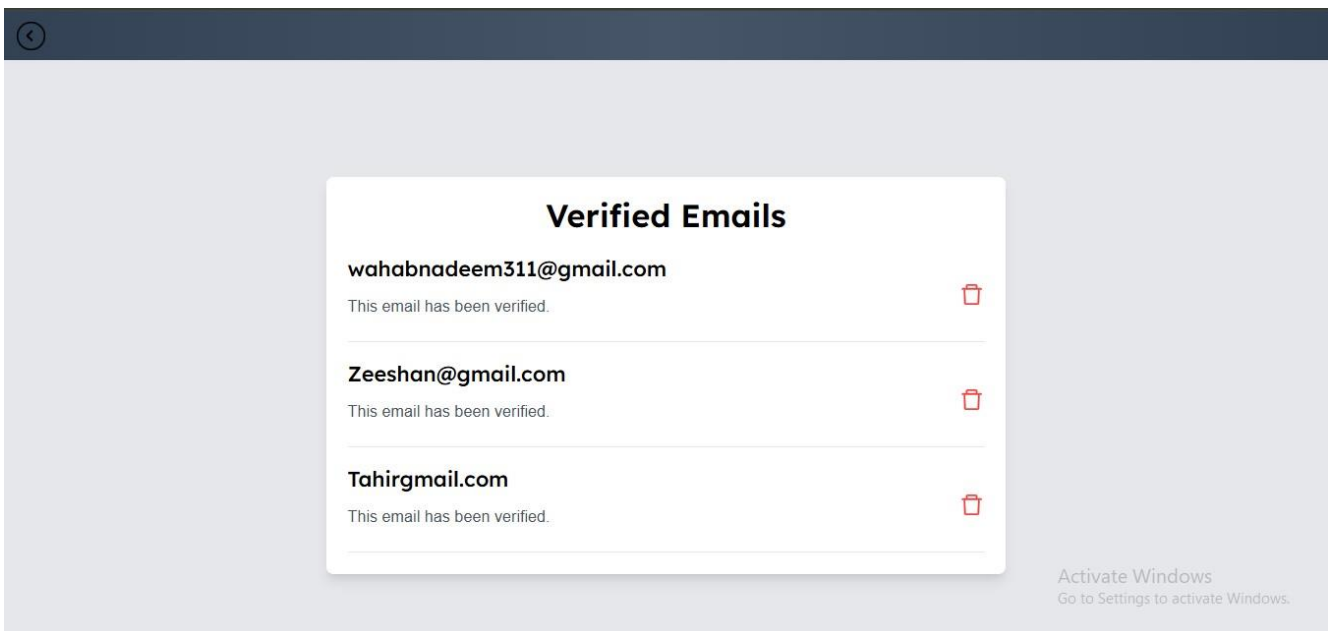
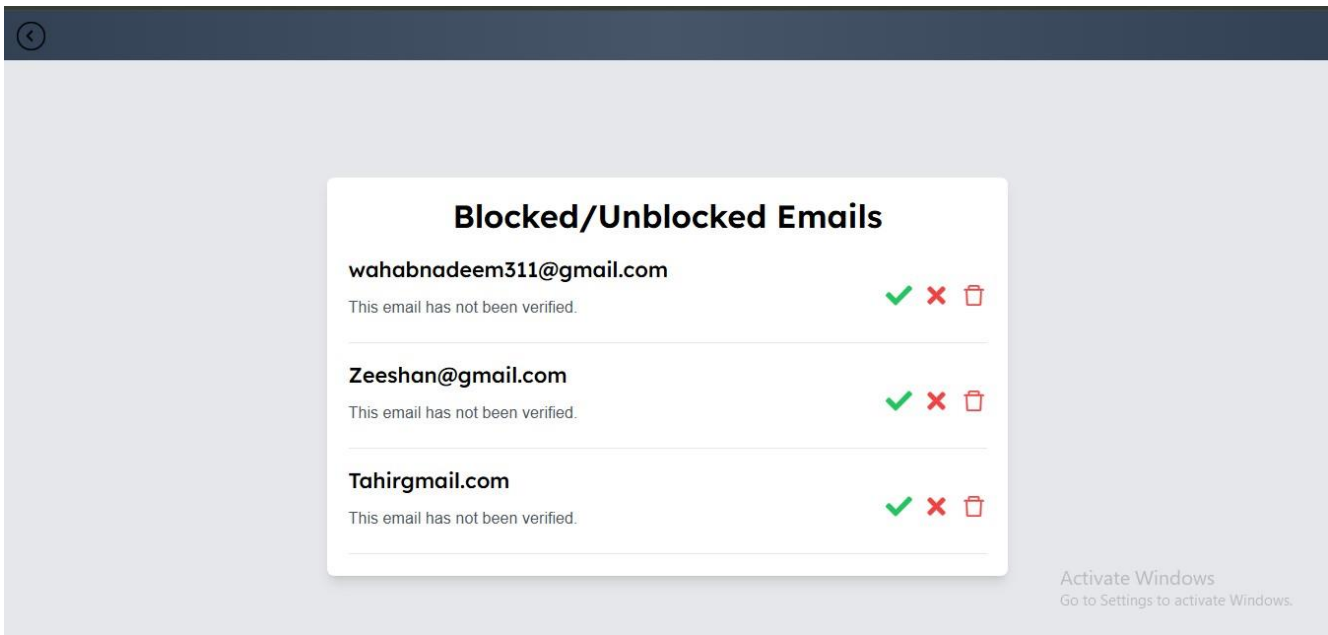
wahabnadeem311@gmail.com

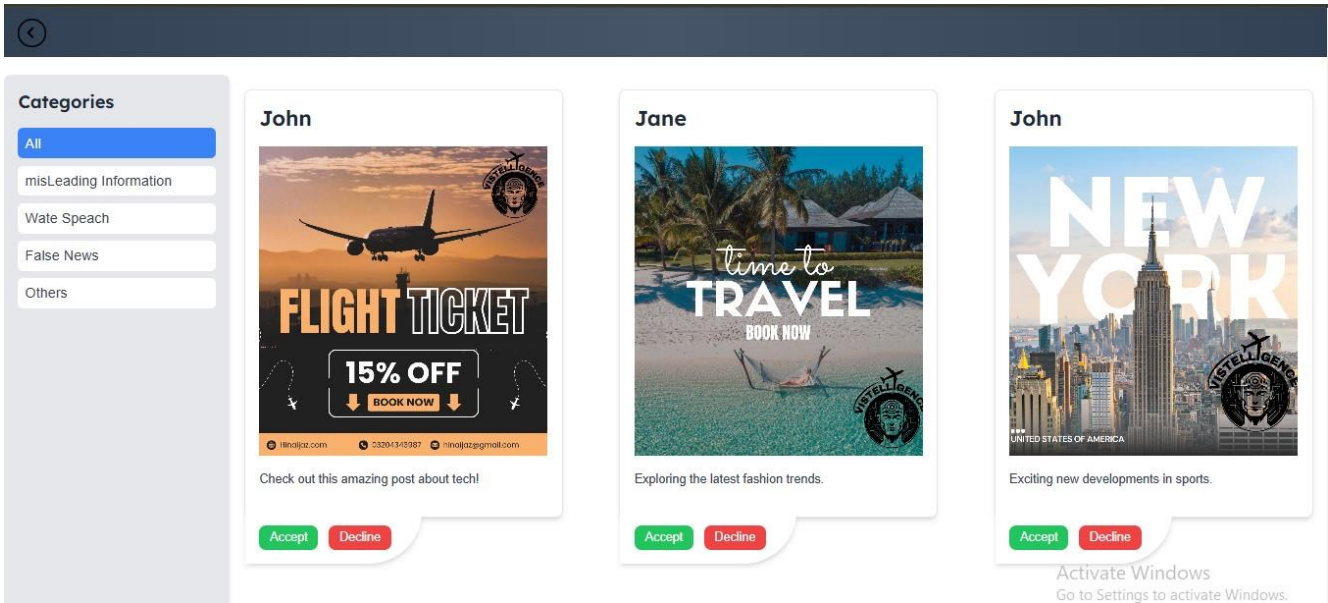
Password

Activate Windows
Go to Settings to activate Windows.



BlockVerifyReported Post





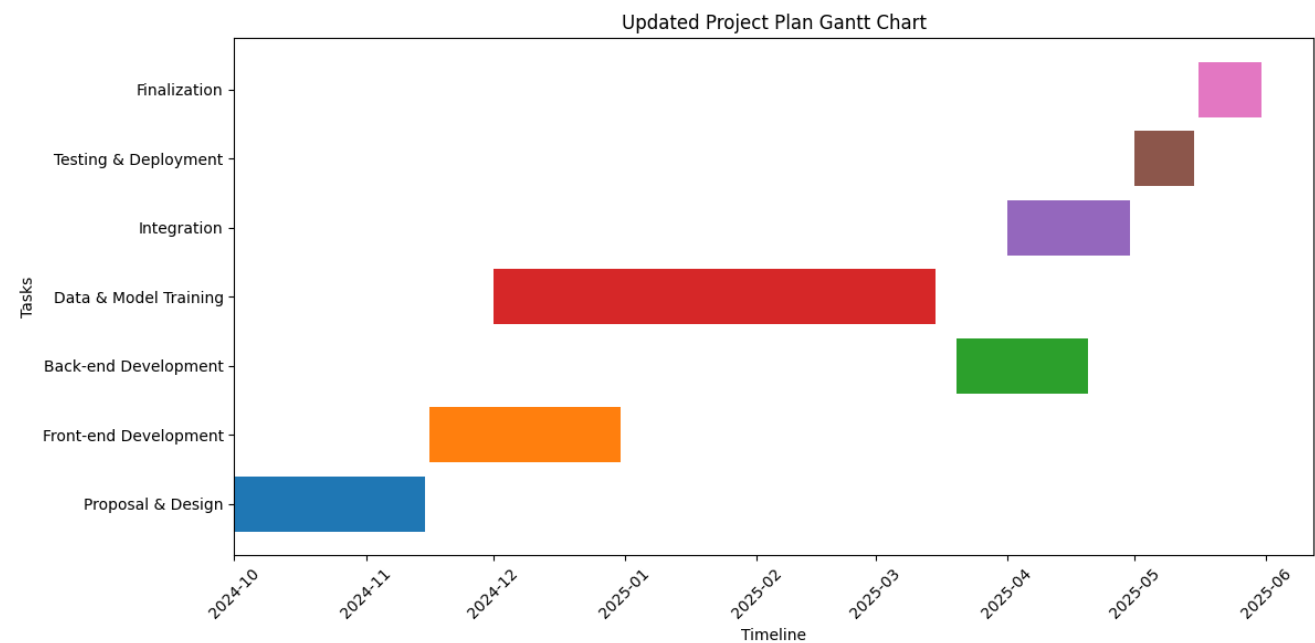
4.3 Additional Information

No Additional Information to share.

5. Other Design Details

No other design details.

6. Revised Project Plan



6.1.1.1 Key Adjustments:

1. **Back-end Development:** Originally scheduled for February, this task has been rescheduled to begin on **20th March** and end on **20th April**.
2. **Integration:** To maintain project continuity, the integration phase has been shifted to start on **1st April** and conclude on **30th April**.

6.1.1.2 Current Progress:

- **Proposal & Design:** Completed as per the initial schedule.
- **Front-end Development:** Ongoing, on track to finish by the original deadline.
- **Data & Model Training:** Progressing as planned, with an expected conclusion in mid-March.
- **Back-end Development:** Scheduled to commence on 20th March.
- **Integration:** Set to begin on 1st April, following the completion of key back-end functionalities.

7. References

ISO/IEC/IEEE 29148:2018. *Systems and Software Engineering — Life Cycle Processes — Requirements Engineering*.

IEEE. (2024). *IEEE Software Engineering Standards*. Retrieved from <https://www.ieee.org/>

Appendix A: Glossary

AI (Artificial Intelligence): Technology used to automate tasks such as visa recommendations, approval predictions, and interview preparation.

- **Approval Chances:** The likelihood of a visa application being approved, based on user data and historical trends.
- **Community Forum:** A space for users to share experiences and discuss the visa application process.
- **Document Checklist:** A list of required documents for specific visa applications.
- **Immigration Advisors:** Experts who assist with visa applications and provide tailored recommendations.
- **Interview Preparation:** Resources and guidance for preparing for visa interviews.
- **MongoDB:** A NoSQL database used for storing user and visa-related data.
- **Personalized Recommendations:** AI-driven suggestions for visa types, countries, and universities based on the user's profile.
- **Profile:** A user's individual information within the platform, used to generate personalized recommendations.
- **Real-Time Updates:** Continuous updates on visa policies and regulations for users.
- **Visa Approval Prediction:** AI-based estimation of the likelihood of visa approval.
- **Visa Types:** Different categories of visas for various purposes (e.g., student, work).
- **Vistelligence:** The AI-powered platform designed to assist with visa applications (our project).
- **User Classes:** Categories of users with different roles and access levels (e.g., students, professionals).
- **User Profile:** The personal information and preferences provided by a user to generate recommendations.
- **Use Case:** A specific function or interaction within the application, such as searching for visa options.
- **Visa Interview Tips:** Guidance to help users prepare for visa interviews.
- **Visa Policy Updates:** Changes to visa regulations and requirements.
- **Visa Strategy:** A personalized plan to improve visa application success.

Appendix B: IV & V Report

(Independent verification & validation)
IV & V Resource

Name _____ Signature _____

S#	Defect Description	Origin Stage	Status	Fix Time	
				Hours	Minutes
1					
2					
3					
...					

Table 1: List of non-trivial defects

This document has been adapted from the following:

- 1. Previous project templates at UCP
- 2. High-level Technical Design, Centers for Medicare & Medicaid Services. (www.cms.gov)

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